

# AI controlled Smart IoT based Greenhouse Monitoring System

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**Abstract**— The AI-controlled Smart IoT-based Greenhouse Monitoring System represents a cutting-edge solution for precision agriculture. Leveraging the power of Artificial Intelligence and the Internet of Things, this system continuously monitors and optimizes environmental variables within the greenhouse, including temperature, humidity, soil moisture, and light levels. Greenhouse gases are the gases in the atmosphere that raise the surface temperature of Earth. Water vapor ( $H_2O$ ), carbon dioxide ( $CO_2$ ), methane ( $CH_4$ ), nitrous oxide ( $N_2O$ ), and Ozone ( $O_3$ ) are some of the examples for greenhouse gases. This article provides an overview of such an automation workflow in greenhouse environments by means of distributed wireless nodes. It makes use of both the IoT and AI technique. Predictions are made through the AI technique which makes use of CNN approach. After a comparative study, sample results from commercial and research greenhouse experiments with the IoT hardware and software have been provided to show connection stability, robustness, and reliability.

**Keywords**— Greenhouse Gases, Internet of Things, Machine-to-Machine (M2M), Wireless Sensor Networks (WSN)

## I. INTRODUCTION

Plant growth thrives under the right balance of environmental conditions. Factors like temperature, humidity, and light play a crucial role in a crop's health and yield. By closely monitoring these variables, growers gain valuable information about their plants' well-being. This knowledge empowers them to fine-tune the growing environment, ultimately maximizing productivity. By harnessing sunlight and manipulating their environment, greenhouses create havens for plant growth. Constructed with translucent materials like glass or plastic, they capture solar energy, fostering a warm and controlled atmosphere. This allows farmers to extend growing seasons, shield crops from harsh conditions, and cultivate delicate or specialized plants. Greenhouses have become cornerstones of modern agriculture, enabling year-round production of a vast array of fruits, vegetables, flowers, and ornamentals. In colder climates, optimizing the greenhouse microclimate is especially important

for maximizing plant output and minimizing energy use. This fine-tuning can significantly boost crop yields and resource efficiency, proving particularly beneficial for agriculture in northern regions.

The Internet of Things (IoT) has transformed greenhouse management by offering real-time monitoring, automation, and data-driven decision-making. A game-changer in this technology is the ability to harness a vast amount of information from strategically placed sensors. These sensors keep a constant eye on critical environmental aspects like temperature, humidity, light levels, soil moisture, and nutrient content. By gathering and interpreting this data, growers can glean valuable understandings about their crops' health and development. This empowers them to make informed choices and fine-tune the growing environment for optimal results. Automation is another important benefit of IoT in greenhouses. Connected devices and actuators can automate processes like altering temperature control systems, irrigation, and fertilizer supply using established parameters or real-time data. For example, IoT-enabled irrigation systems may direct the appropriate amount of water to plants depending on soil moisture levels, eliminating water waste and maintaining adequate hydration. Remote monitoring and control are also facilitated by IoT technology. Farmers can access real-time data and control greenhouse operations from anywhere using smartphones, tablets, or computers connected to the internet. This capability enables quick responses to changing conditions, remote troubleshooting, and efficient resource management. Compared to wired systems, Wireless Sensor Networks (WSNs) boast several advantages, making them a game-changer in the age of the Internet of Things (IoT) and Machine-to-Machine (M2M) communication. These benefits include high precision data collection, the ability to function even with some sensor failures, easy deployment in diverse settings, affordability, self-sufficiency in terms of power, and resilience in harsh environments. As a result, WSNs are rapidly becoming an essential technology in various fields, including military operations, environmental monitoring, healthcare ap-

plications, smart home automation, and numerous commercial ventures.

The Internet of Things (IoT) connects a vast array of devices, enabling them to communicate and interact with each other or with humans. A prime example of this in agriculture is the node-based greenhouse system. This system leverages strategically placed sensors to create a real-time network that monitors the intricate details of the greenhouse environment. This constant stream of data provides farmers with crucial insights into the factors that influence plant growth, empowering them to make informed decisions for optimal cultivation.

Greenhouse management gets a boost from Artificial Intelligence (AI). AI's sophisticated algorithms unlock a new level of optimization, leading to bigger yields and smarter resource use. A key feature is AI's ability to analyze mountains of data. This data can come from greenhouse sensors, weather forecasts, and even past growing seasons. By crunching these numbers, AI provides real-time insights and empowers growers to make the best choice for their crops.

## II. RELATED WORKS

The literature survey explores a spectrum of methodologies in image caption generation, each offering unique insights and advancements. This article introduces a prototype wireless sensor network for greenhouses. MicaZ nodes, known for low-power communication, measure key environmental factors like temperature and humidity. Farmers can monitor these conditions remotely using smartphones or computers with an internet connection. The current system allows for data collection and analysis, but future plans include incorporating actuators for two-way communication. This will enable remote control of the greenhouse environment, creating a more precise and automated system for optimal plant growth. [1]

This literature [2] discuss about use of IoT technology in agriculture, with a focus on greenhouse applications. It includes a systematic review of relevant scientific articles and highlights the benefits and potential applications of this technology.[2]

The research article[3] discusses how machine learning can be used to increase the productivity of agriculture through approaches as a fully automatic greenhouse that uses different environmental sensors that record timely atmospheric conditions and mechanical actuators that replace the labour intensive achievement of maintaining meticulous atmospheric conditions in a confined room.

Early disease detection, reduced crop loss. Explore multi-sensor fusion for comprehensive disease detection. Image processing, machine learning models. Implement machine learning for early detection of crop diseases in smart Greenhouses.[4] Moreover, The literature survey contains references on various topics such as optimal control of greenhouse cultivation, path planning in dynamic environments, and sample-efficient reinforcement learning via model ensemble exploration and exploitation. It also includes articles on the use of digital technologies in agriculture and neural network predictive control in a naturally ventilated and fog-cooled greenhouse [5]

Automates monitoring and empowers farmers with valuable insights for informed decision-making. The AI algorithms are trained on historical and real-time data collected from various sensors deployed in the greenhouse, including temperature, humidity, moisture, and light intensity sensors. Dynamically adjusts environmental parameters based on AI predictions, the system enhances the overall efficiency of greenhouse operations.[6]

The literature [7] survey discusses an Internet of Things based automated smart greenhouse system using adaptive neuro- fuzzy inference system (ANFIS) for optimizing weather parameters, collecting and visualizing data from various sensors in real time, and ensuring data security through identifying potential IoT perceptron layer attacks and taking necessary steps to prohibit them. The network is fabricated to boost crop production and provide data traceability and durability for customized indoor agriculture quality and safety.

Furthermore, the literature [8] aimed at identifying the main information about the Internet of Things in Smart Greenhouse (IoTSG) that was documented. The SLR process was carried out methodically and systematically to produce a review that can be imitated by other researchers, and to evaluate all existing evidence on a validated research topic. The SLR found that the general number of sensors used in IoTSG was 26, and the most widely used sensors were temperature and humidity sensors with 12, followed by soil moisture sensors.

The presented [9] paper discuss the MQTT protocol and its different Quality of Service (QoS) levels as well as solutions to ensure no information loss, the analysis of IoT evolution, secure communication in IoT devices, and the need for unique IoT administration at home. The proposed Adaptive On Demand Transmission Power Control protocol for the greenhouse application is also discussed, along with data acquisition and logging of soil pH level and temperature in a greenhouse environment.

Dynamically adapt to changing environmental conditions. It gives improved yield consistency and quality. The study involves a comparative analysis, evaluating the performance of the AI-controlled system against traditional greenhouse management methods[10]

The literature[11] primarily discusses a greenhouse monitoring system with remote control and automatic data acquisition capabilities. It includes modules for monitoring greenhouse environments, remote control of equipment, data acquisition and storage, and system management. Specific topics covered are the system structure, software and hardware design, data transmission and storage, and user interface design. This literature survey shows that previous work on automation only pertains to controller-made decisions. Other researchers have used M2M communication, image processing, and decision tree algorithms to improve irrigation efficiency. F.P. Jooste et al. implemented an irrigation system focused on energy efficiency[12]. The system leverages sensors and an Arduino board, powered by solar panels, to monitor and regulate key environmental factors. Uniquely, farmers can access real-time data from anywhere through a custom web interface, Android app, or on-site LCD display. This innovation aims to address the challenges faced by farmers in maintaining optimal growing conditions for their crops[13].

The literature[14] discusses various features of artificial intelligence-enabled smart monitoring and also controlling of IoT-greenhouses, including adaptive activation functions in deep neural networks, time series prediction using LSTM neural networks for smart agriculture applications, and efficient optimization techniques for LSTM neural networks, among others. The survey also covers implementation of Laravel framework in web-based temperature and also humidity monitoring systems and the survey techniques of artificial intelligence over field of Internet of Things.

The literature [15,16] provides a critique of recent analysis on smart greenhouses and artificial intelligence technologies, identifying both favourable factors and challenges for the growth of these technologies. It highlights the benefits of smart farming technologies, mechanization of farms, smart micro-irrigation, rapid soil analysis, and pest prediction in improving farming efficiency and productivity. The review also emphasizes the potential of LIDAR, optical and electrooptical cameras for robotic system motion and object identification, and the combination of AI and IoT to address challenges related to global food insecurities and climate change.

### III. METHODOLOGY

In this methodology section, we present the comprehensive framework for our image captioning project.

#### A. Architecture System

The system takes the readings from the 3 basic sensors : temperature, moisture and humidity These readings will be monitored from the app which is communicated through the real time database 'Firebase', this is all done through wifi control. Firebase takes the real-time readings from the sensors, then this will be depicted in a Mobile application. Suppose if a hot environment, then the cooling fans would get ON. If a cool environment inside the greenhouse then the LED lights will be ON. If soil that is dry, then system will automate the solenoid valve for water, all these cooling fans, heat generators or solenoid valve are automated, i.e they will work automatically. A camera is installed inside the greenhouse to take pictures and for classifying whether it's condition is good or bad through the Wi-Fi control which will be sent to real-time database. Fig. 1 depicts the following example.

#### B. Process Flow

Technology is used by a greenhouse temperature control system to provide the perfect growing environment for plants. This diagram shows a greenhouse where several features are managed by a central control unit that could be a computer. Sensors keep an eye on the humidity, accurate temperature, and probably light levels. In order to maintain stable conditions, the control unit then regulates artificial lighting, humidity control systems, and heating. Finest plant growth throughout the network is ensured via remotely monitoring and changes made possible by an IoT gateway. Adding an IoT gateway raises the bar for sophistication. It basically establishes a network between the system and the internet, enabling remote monitoring and control.

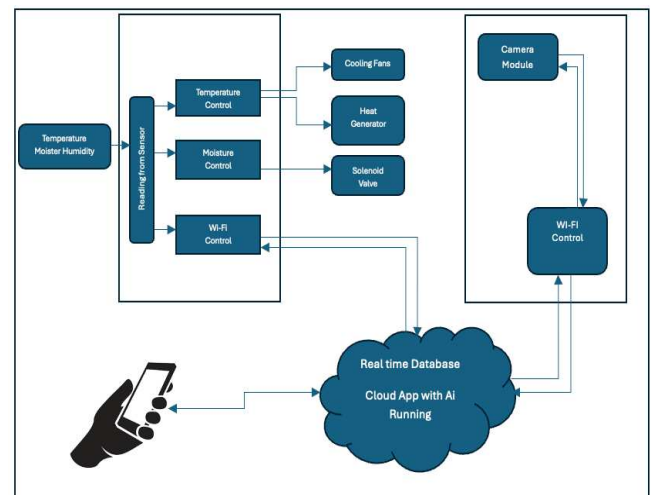


Fig. 1. Architecture diagram for the proposed module

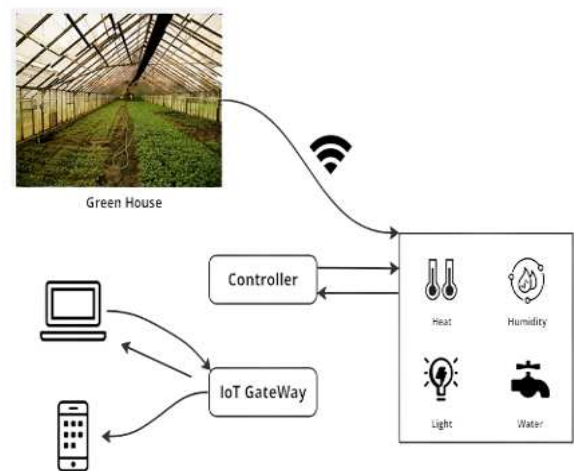


Fig. 2. A process flow for the proposed module

Imagine being able to monitor the weather conditions of your greenhouse from anywhere with a computer or smartphone.

This remote access allows for quick intervention if needed and ensures finest growing conditions are maintained throughout the greenhouse, fostering healthy and thriving plants. Fig. 2 represents the following module.

#### C. Greenhouse System

A Greenhouse represents the overall controlling and monitoring system, and the greenhouse network is a supervised environment used for growing plants, particularly those that require specific conditions to thrive. It consists of structure made of see-through material, such as glass or plastic, that allows sunlight to enter while trapping heat inside, creating a warmer climate than the surrounding outdoor environment.

#### D. Sensor Module

Tiny but mighty, sensor modules are electronic packages that detect and measure the world around them. They come equipped with sensors like temperature gauges, or light

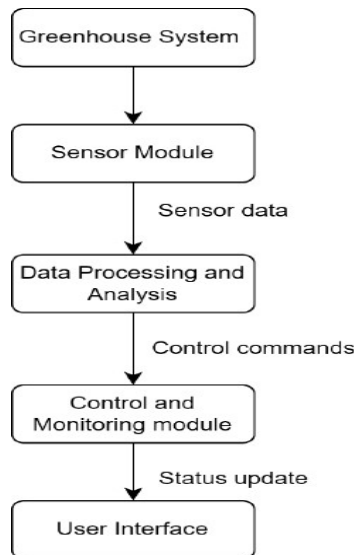


Fig. 3. A block diagram of working of greenhouse system

detectors, depending on their purpose. These sensors gather environmental data and convert it into electrical signals for microcontrollers or other electronic systems to understand. Sensor modules are like the eyes and ears of various fields: automation, robotics, environmental monitoring, healthcare, and even smart homes. They provide real-time data, allowing for better decision-making, control, and optimization. By simplifying data collection, sensor modules play a key role in the Internet of Things (IoT), making it possible to monitor and manage a vast array of systems and applications.

#### E. Data Processing and Analysis

Data processing and analysis refers an approach of converting raw data into meaningful information through various computational and statistical techniques. Processes the incoming sensor data, performs analysis, and generates control commands based on predefined rules or user settings. This may involve duties like data cleaning to remove errors and inconsistencies, data integration to combine data from multiple sources, and data transformation to standardize formats or calculate derived variables.

#### F. Control and Monitoring Module

The brains behind automation, control and monitoring modules keep various systems running smoothly. These modules act as a central nervous system, overseeing processes and devices. They typically use sensors to gather real-time data, microcontrollers to analyze it, and actuators to adjust parameters based on those findings. By constantly monitoring the system's performance and comparing it to desired settings, these modules can identify deviations and take corrective actions. This ensures the system operates efficiently, safely, and reliably within defined standards. Common applications include automation, industrial processes, environmental controls, smart buildings, and the Internet of Things (IoT). Control and monitoring modules play a critical role in optimizing performance, minimizing downtime, and enhancing overall system functionality.

Control and monitoring module also use CNN based leaf disease detection.

#### G. User Interface

Interacts with users, displays real-time data, allow them to set preferences, and receives status updates from the control as well as monitor module.

### IV. EXPERIMENTATION AND RESULTS

An AI-powered greenhouse system takes precision agriculture to new heights. Sensors monitor resource use, and AI optimizes distribution to minimize waste and costs. It also predicts crop yields and analyzes energy use to promote sustainability. AI even detects potential problems early using sensor data, allowing farmers to take swift action. Finally, a mobile app gives growers real-time data for informed decisions.

Smart greenhouses powered by AI and the Internet of Things (IoT) are revolutionizing agriculture by significantly boosting productivity. Real-time data analysis from strategically placed sensors, combined with powerful AI algorithms, allows for precise control over environmental factors. This creates optimal growing conditions for plants, leading to higher yields. The system's predictive capabilities empower farmers to proactively manage potential problems before they arise, minimizing crop losses and maximizing overall output. Additionally, seamless communication between IoT sensors ensures efficient resource utilization, reducing waste and promoting sustainable practices. The integration of artificial intelligence not only automates routine tasks but also provides valuable insights through data analysis, enabling informed decision-making for farmers. In essence, this technology represents a pivotal advancement in modern agriculture, fostering a more intelligent and resource-conscious approach to greenhouse cultivation. All components are initialized by delivering the necessary electricity. Three sensors, humidity, temperature, and soil moisture are designed to be employed. The Arduino is connected to a cooling fan, a water pump, an LED light source, and a motor pump.

Ethernet is utilized to communicate environmental characteristics with the web interface All environmental parameters are relayed to the server via Ethernet. High temperatures frequently cause crop failure and a decline in fruit yield.

High humidity affects plant photosynthesis, transpiration processes, and nutrient absorption. An overheated greenhouse system causes the soil to quickly dry up, requiring more water to maintain moisture. When the temperature exceeds a certain level, the system turns on the fan which reduces the system inside and when the temperature level is at normal range the fan turns off by itself making it completely automated to use. Moisture sensor monitors the moisture content of the soil. Water is an essential component of plant growth. When the moisture sensor does not detect moisture in the soil, the system activates the motor pump, which supplies water and causes the soil to become moist.

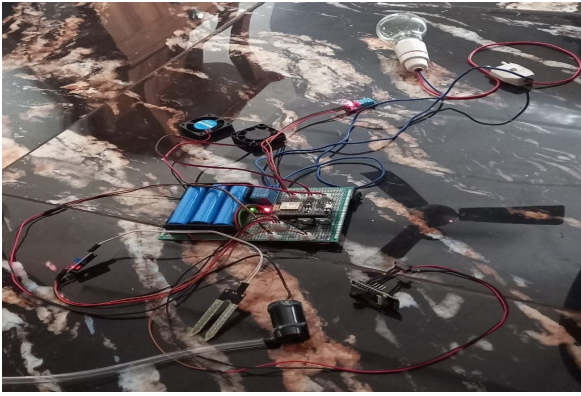


Fig. 4. A block diagram of working of greenhouse system



Fig. 5. Arduino Monitor Output

When the moisture in the soil reaches a normal stated range, the motor pump is turned off as showed in the Fig. 4. All the values that are measured from the ESP-12E Module are sent as outputs to the monitor screen through which the results can be monitored and visualized. The outputs from the Temperature, Humidity and Soil moisture sensors are accurate and regularly varying according to the environmental parameters of the greenhouse as showed in the Fig. 5.

This line graph of temperature being varied over the time with data obtained through real time-tested values. This graph shows a time series plot with temperature on the y-axis and time on the x-axis. The temperature is in degrees Celsius ( $^{\circ}\text{C}$ ). From the graph we can analyze that here the system allows till the temperature to reach to maximum threshold point till 35.5 degree Celsius and later the temperature is made to cool down by using fan.

The graph shows a humidity over time graph on which humidity on y- axis and time on the x-axis. The humidity scale goes from 69.0% to 70.0% in most of the times.

## V. CONCLUSION

This article utilizes an Arduino microcontroller to create a greenhouse monitoring and control system. Sensors for temperature, humidity, moisture content, and light intensity provide real-time data on the greenhouse environment.

A web interface allows for control as well as monitoring of these environmental parameters.

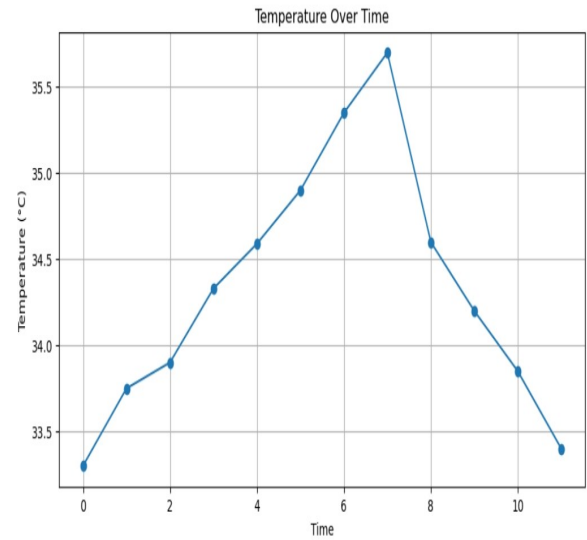


Fig. 6. A plot of temperature change along with time in greenhouse system

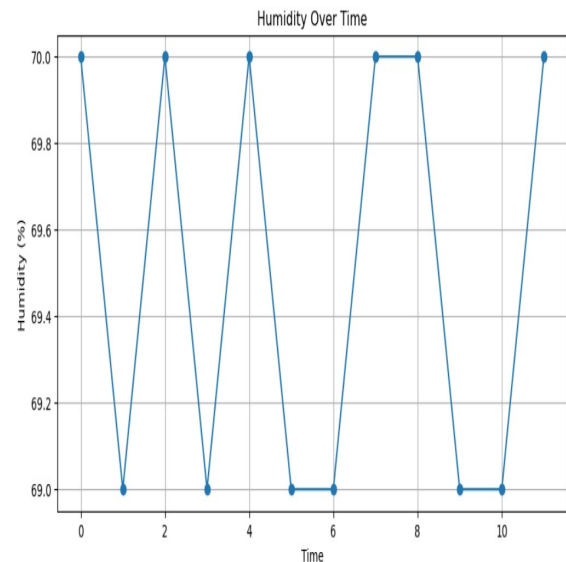


Fig. 7. A plot of changes in humidity with time in greenhouse system

Additionally, the system leverages ethernet to transmit data to mobile phones, eliminating SMS costs. This approach reduces power consumption, maintenance needs, and overall system complexity. The project's applications extend to various agricultural settings, including fields, nurseries, and botanical gardens.

## VI. ACKNOWLEDGEMENTS

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